STAT/MA 41600
In-Class Problem Set #31: October 28, 2026
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Problem Set 31 Answers

1a. The lines $y = 5 - 5x/8$ and $y = x$ meet at the point where $x = y = 40/13 = 3.0769$. There are (at least) two ways to find $P(Y > X)$. One way is to calculate the ratio of the areas, $\frac{(1/2)(40/13)(5)}{(1/2)(8)(5)} = 5/13$. Another way is to compute $\int_0^{40/13} \int_x^{5-5x/8} 1/20 \, dy \, dx = 5/13$.

1b. There are (at least) three ways to find $P(X + Y \geq 3)$. One way is to calculate the ratio of the areas, $\frac{(1/2)(8)(5) - (1/2)(3)^2}{(1/2)(8)(5)} = 31/40$. Another way is to use the complement: $1 - \int_0^3 \int_0^{3-x} 1/20 \, dy \, dx = 31/40$. A more challenging way is to compute $\int_0^3 \int_{3-x}^{5-5x/8} 1/20 \, dy \, dx + \int_3^8 \int_0^{5-5x/8} 1/20 \, dy \, dx = 123/320 + 25/64 = 31/40$.

2a. For $0 < a < 10$, we have $P(Y \leq a) = P(\max(X_1, \ldots, X_n) \leq a) = (a/10)^n$, where the last equality holds since $X_1, \ldots, X_n$ are independent. So we have:

$$F_Y(y) = \begin{cases} 0 & \text{if } y \leq 0, \\ (y/10)^n & \text{if } 0 < y < 10, \\ 1 & \text{if } y \geq 10. \end{cases}$$

2b. For $0 < y < 10$, we have $f_Y(y) = \frac{d}{dy} F_Y(y) = n(y/10)^{n-1}(1/10)$, and otherwise $f_Y(y) = 0$.

3ab. Yes, we have $f_X(x) = \frac{d}{dx} F_X(x) = 1/40$ for $-20 < x < 20$, and otherwise $f_X(x) = 0$. So X is a continuous uniform random variable, namely, uniformly distributed on $[-20, 20]$.

3c. We have $E(X) = \int_{-20}^{20} (x)(1/40) \, dx = 0$.

3d. We have $E(X^2) = \int_{-20}^{20} (x^2)(1/40) \, dx = 400/3$, so $\text{Var}(X) = E(X^2) - (E(X))^2 = 400/3 - 0^2 = 400/3$.

4ab. We note Y is distributed on $[-20a+b, 20a+b]$; for y in this interval, we have $F_Y(y) = P(Y \leq y) = P(aX + b \leq y) = P(X \leq (y-b)/a) = F_X((y-b)/a) = (y-b)/(40a) + 1/2$, so $f_Y(y) = \frac{d}{dy} F_Y(y) = 1/(40a)$; otherwise, we have $f_Y(y) = 0$. Therefore Y is a continuous uniform random variable, namely, uniformly distributed on $[-20a + b, 20a + b]$.

4c. We have $E(Y) = E(aX + b) = aE(X) + b = b$. Alternatively, we can calculate $E(Y) = \int_{-20a+b}^{20a+b} (y)(1/(40a)) \, dy = b$.

4d. We have $\text{Var}(Y) = \text{Var}(aX + b) = a^2 \, \text{Var}(X) = (a^2)(400/3)$. Alternatively, we can calculate $E(Y^2) = \int_{-20a+b}^{20a+b} (y^2)(1/(40a)) \, dy = (a^2)(400/3) + b^2$, and it follows that $\text{Var}(Y) = E(Y^2) - (E(Y))^2 = (a^2)(400/3)$.

5. We compute $P(X > Y) = \int_0^2 \int_0^x (1/2)(7e^{-7y}) \, dy \, dx = (13 + e^{-14})/14$.

Alternatively, we can compute $P(X > Y) = \int_0^2 \int_y^2 (1/2)(7e^{-7y}) \, dx \, dy = (13 + e^{-14})/14$.